Applications of Abstract Algebra in Numerical Analysis: Polynomials in Lagrange Interpolation

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1 Introduction

Polynomials are one of the major abstract algebra ideas that is commonly used in high school and college-level mathematics. Not only are polynomials used in elementary algebra and calculus, but the ideas and concepts of polynomials as well play a significant role in other areas applied mathematics. One such area of applied mathematics in which the principles of polynomials play a significant role is numerical analysis. Polynomials are associated with polynomial interpolation, which is a technique used to estimate a certain value of a function using polynomials and the function's given values.

Some numerical analysis interpolation methods involve polynomials, such as Lagrange interpolation, Hermite interpolation, and so on. In this paper, we will look specifically at the definition of Lagrange interpolation and how abstract algebra theorems on polynomial concepts can be used to find and prove the properties of a Lagrange polynomial. But before we go deeper into applying the theorems into Lagrange interpolation, we must first look at the definition of a polynomial and its properties.

2 Polynomials in Abstract Algebra

Polynomials are denoted by $p(x) = c_0 + c_1x + \ldots + c_nx^n$, where c_ix^i and $(0 \leq i \leq n)$ are called the terms of the polynomial and $c_0, c_1, \ldots, c_n$ are the coefficients of $p(x)$. In addition, we say that n is the degree of polynomial if n has the greatest value such that the term x^n is non-zero. In A Book of Abstract Algebra, authored by Charles Pinter, the polynomial $p(x)$ is said to be over field F, which means that the coefficients in the polynomial belong to F. This makes sense because the coefficients are commutative under addition (which tells us that there exists an additive identity 0 and an additive inverse element) and they are also commutative under multiplication. Moreover, the coefficients are associative under multiplication and there exists a multiplicative identity 1, as well as that multiplication is distributive over addition and that all non-zero coefficients are invertible (according to class notes for fields and special rings).

It is important to understand that polynomials do not belong to a field because non-zero polynomials are not invertible, as taking the multiplicative (but not functional) inverse of a polynomial will not result in a polynomial. However, by Theorem 1 in Chapter 24, polynomials belong to a commutative ring with unity $A(x)$ if A is a commutative ring with unity and consists of the coefficients of the polynomial (Pinter 243). If there exist a $k \in F$, we can substitute k for x in the polynomial $p(x)$ to obtain $p(k) = c_0 + c_1k + \cdots + c_nk^n$ and that $p(k)$ belongs to field F as well (Pinter 258).

Nevertheless, Pinter wants us to consider $p(x)$ to be a polynomial over a field so that the following theorems takes place; Theorem 1 in the book states that "c is a root of $p(x)$ if and only if $x - c$ is a factor of $p(x)$" and Theorem 2 tells us that "if $p(x)$ has distinct roots $c_1, \ldots, c_m$ in F, then $(x-c_1)\cdot(x-c_2)\cdot\ldots\cdot(x-c_m)$ is a factor of $p(x)$" (Pinter 259). These concepts about factors, roots, and substitution in polynomials play a big role in proving the properties of Lagrange interpolation and will be used in Lagrange interpolation.

3 Lagrange Interpolation

According to Pinter's book, which is discussed in Chapter 26 Exercise H, Lagrange interpo-

lation focuses on finding a polynomial p with degree at most n that interpolates a set of given data points (x, y) for $x = a_0, a_1, \ldots, a_2$ and $y = b_0, b_1, \ldots, b_n$ (268) or a function $f(x)$ with given x values such that $x = a_0, a_1, \ldots, a_n$. In addition, that polynomial satisfies $y = p(x)$ such that $p(a_0) = b_0, p(a_1) = b_1, \ldots, p(a_n) = b_n$.

Now, suppose that $q_i(x) = (x - a_0) \cdot \ldots \cdot (x - a_{i-1}) \cdot (x - a_{i+1}) \cdot \ldots \cdot (x - a_n)$, where $i = 0, 1, \ldots, n$ and define $q_i(a_i) = c_i$ which implies that $c_i = (a_i - a_0) \cdot \ldots \cdot (a_i - a_{i-1}) \cdot (a_i - a_{i+1}) \cdot \ldots \cdot (a_i - a_n)$. The interpolating polynomial p(x) is defined as follows:

$$\begin{aligned} p(x) &= \sum_{i=0}^{n} \frac{b_i}{c_i} \cdot q_i(x) \\ &= \frac{b_0}{c_0} \cdot q_0(x) + \ldots + \frac{b_n}{c_n} \cdot q_n(x) \end{aligned}$$

Where $q_i(x) = (x - a_0) \cdot \ldots \cdot (x - a_{i-1}) \cdot (x - a_{i+1}) \cdot \ldots \cdot (x - a_n)$ and this is called the Lagrange interpolation formula (Pinter 268).

In contrast, according to Numerical Analysis: Mathematics of Scientific Computing book authored by David Kincaid, the Lagrange interpolation formula has the same polynomial but has a different form as follows:

$$\begin{aligned} p(x) &= \sum_{k=0}^{n} y_k \cdot l_k(x) \\ &= y_0 \cdot l_0(x) + y_1 \cdot l_1(x) + \ldots + y_n \cdot l_n(x) \end{aligned}$$

Where l_k is defined to be a cardinal function and has the form

$$l_k(x) = \prod_{j=0, j \neq k}^{n} \frac{x - x_i}{x_k - x_j}$$

given that (x_k, y_k) are the data points used to find the polynomial (Kincaid 312).

4 Factors and Substitutions in Polynomials and Lagrange Interpolation Relation

First, we want to show why $p(a_0) = b_0, p(a_1) = b_1, \ldots, p(a_n) = b_n$ using Lagrange interpolation formula and the concept of substitution in polynomials. Pinter states that if there exists a $c \in F$, we can substitute the value of c for x in the polynomial $q_i(x)$ and it is denoted by $q_i(c)$[U+2008] $= (c - a_0) \ldots (c - a_{i-1})(c - a_{i+1}) \ldots (c - a_n)$ and that $q_i(c)$ is an element of F, therefore $q_i(x)$ is a function from F to F called a polynomial function on F (258).

Here, let us substitute a_0 for x in $p(x)$, then we have

$$\begin{aligned} p(a_0) &= \sum_{i=0}^{n} \frac{b_i}{c_i} \cdot q_i(a_0) \\ &= \frac{b_0}{c_0} \cdot q_0(a_0) + \ldots + \frac{b_n}{c_n} \cdot q_n(a_0) \end{aligned}$$

Notice that $x = a_0, \ldots, a_{i-1}, a_{i+1}, \ldots, a_n$ are the distinct roots of $q_i(x)$ because according to Theorem 1 which states that "c is a root of $a(x)$ if and only if $x - c$ is a factor of $a(x)$" (Pinter 259), $x - a_0, \ldots, x - a_{i-1}, x - a_{i+1}, \ldots$, and $x - a_n$ are the factors of $q_i(x)$ since $q_i(x)$ can be written as the product of all $x - a_0, \ldots, x - a_{i-1}, x - a_{i+1}, \ldots$, and $x - a_n$ that looks like the division algorithm of a polynomial $q_i(x) = (x - a_j)k(x) + r(x)$ (Pinter 245), where $k(x)$ is the product of all other $x - a_k$ for $k \neq j$ and $r(x) = 0$. However, $x - a_i$ is excluded which means that $x = a_i$ is not a root of $q_i(x)$ and therefore, given that

$$\begin{aligned} q_i(x) = (x - a_0) \cdot \ldots \cdot (x - a_{i-1}) \\ \cdot (x - a_{i+1}) \cdot \ldots \cdot (x - a_n) \end{aligned}$$

Then, $q_i(a_i) \neq 0$ and for each $j \neq i$, we have $q_i(a_j) = 0$ since each a_j is a root of $q_i(x)$ (Pinter 268). Since we know that $q_i(a_j) = 0$ for $j \neq i$ and as we have defined $q_i(a_i) = c_i$, these imply that $q_1(a_0) = q_2(a_0) = \ldots = q_n(a_0) = 0$ and that $q_0(a_0) = c_0$. Therefore, using the substitution method that Pinter stated in chapter 26, we have

$$\begin{aligned} p(a_0) &= \sum_{i=0}^{n} \frac{b_i}{c_i} \cdot q_i(a_0) \\ &= \frac{b_0}{c_0} \cdot c_0 + \frac{b_1}{c_1} \cdot 0 + \ldots + \frac{b_n}{c_n} \cdot 0 \end{aligned}$$

and as a result, we have $p(a_0) = \frac{b_0}{c_0} \cdot c_0 = b_0$. When we substitute a_1 for x, we have $q_1(a_1) = c_1$ and $q_0(a_1) = q_2(a_1) = \cdots = q_n(a_1)$, so we obtain

$$\begin{aligned} p(a_1) &= \sum_{i=0}^{n} \frac{b_i}{c_i} \cdot q_i(a_1) \\ &= \frac{b_0}{c_0} \cdot 0 + \frac{b_1}{c_1} \cdot c_1 + \ldots + \frac{b_n}{c_n} \cdot 0 \end{aligned}$$

which leads us to $p(a_1) = \frac{b_1}{c_1} \cdot c_1 = b_1$ and this implies that $p(a_2) = b_2, \ldots, p(a_n) = b_n$, so we come to a conclusion that the interpolating Lagrange polynomial satisfies $p(a_0) = b_0, p(a_1) = b_1, \ldots, p(a_n) = b_n$ (Pinter 268).

It is also important to know that the interpolating Lagrange polynomial of degree at most n such that $p(a_1) = b_1, \ldots, p(a_n) = b_n$ is unique (Pinter 268). If there exist two polynomials, say $p(x)$ and $q(x)$, and these satisfy the given condition such that

$$p(a_0) = b_0, p(a_1) = b_1, \ldots, p(a_n) = b_n$$

$$q(a_0) = b_0, q(a_1) = b_1, \ldots, q(a_n) = b_n$$

This implies that $p(a_0) = q(a_0), p(a_1) = q(a_0), \ldots, p(a_n) = q(a_n)$ and since $a_0, \ldots, a_n \in x$, we have $p(x) = q(x)$ and therefore the Lagrange interpolating polynomial $p(x)$ is unique.

We can also show the uniqueness of the Lagrange interpolating polynomial by letting there exist two polynomials $p(x)$ and $q(x)$ of degree n which interpolate the same data point (x, y), this implies that the polynomial $p(x) - q(x)$ is of degree n. According to Theorem 3, a polynomial $p(x)$ has at most n roots if it has degree n (Pinter 260) , but since $x = a_0, a_1, \ldots, a_n$ are distinct, the polynomial $p(x) - q(x)$ has $n+1$ roots/zeros, and therefore $p(x) - q(x)$ must be a zero polynomial (Kincaid 309) meaning that $p(x) - q(x) = 0$ which implies $p(x) = q(x)$. Therefore, the Lagrange interpolating polynomial is unique (Pinter 268).

Example

According to MATH 450 Module 3 Notes, let there be three data points $(1, 1), (2, 3)$, and $(4, 3)$ We know that $a_0 = 1, a_1 = 2, a_2 = 4$ and $b_0 = 1, b_1 = 3, b_2 = 3$, now we need to find $\frac{q_i(x)}{c_i}$. Using the substitution in polynomials method that is explained in Pinter's book (258), we obtain:

$$\frac{q_0(x)}{c_0} = \frac{(x-2)(x-4)}{(1-2)(1-4)} = \frac{1}{3}(x-2)(x-4)$$

$$\frac{q_1(x)}{c_1} = \frac{(x-1)(x-4)}{(2-1)(2-4)} = -\frac{1}{2}(x-1)(x-4)$$

$$\frac{q_2(x)}{c_2} = \frac{(x-1)(x-2)}{(4-1)(4-2)} = \frac{1}{6}(x-1)(x-2)$$

Hence, we have the Lagrange interpolating polynomial $p(x)$ of form:

$$\begin{aligned} p(x) &= \sum_{i=0}^{2} \frac{b_i}{c_i} \cdot q_i(x) \\ &= \frac{b_0}{c_0} \cdot q_0(x) + \frac{b_1}{c_1} \cdot q_1(x)s + \frac{b_2}{c_2} \cdot q_2(x) \end{aligned}$$

Using the substitution in polynomials (Pinter 258) and the fact that polynomials belong to a commutative ring with unity $A(x)$ (Pinter 243), we can distribute the polynomial form and obtain:

$$\begin{aligned} p(x) &= \frac{1}{3}(x-2)(x-4) - \frac{3}{2}(x-1)(x-4) \\ &+ \frac{1}{2}(x-1)(x-2) \\ &= -\frac{2}{3}x^2 + 4x - \frac{7}{3} \end{aligned}$$

Where $-2/3, 4$, and $-7/3$ belong to a commutative ring with unity A.

5 Conclusion

Polynomials are one of the simplest functions to compute, and because they have so many applications in real life (Pinter 240), they must play a significant role in many areas of mathematics. Not just computing involved in polynomials, it is also important to understand the properties and why these make sense. Furthermore, in applied mathematics, we deal with some complicated functions, and we can express the complicated function in terms of polynomials thanks to abstract algebra polynomials and Lagrange interpolation. Therefore, abstract algebra polynomial concepts such as substitution, factors, roots, and operations are applicable to Lagrange polynomials.

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7 References

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